

1 **Focused-Universe Reanalysis of Complement, Coagulation, and Extracellular-Matrix**

2 **Transcriptional Programs in Diabetic Kidney Disease**

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8 **Abstract**

9 Outcome-dependent gene filtering can invalidate conventional confirmation claims when
10 discovery datasets are reused in meta-analysis. We reanalyzed nine biologically defined gene
11 sets comprising 145 unique genes in human diabetic kidney disease transcriptomes. Discovery
12 direction was estimated from GSE142025 and GSE96804. A selection-independent three-study
13 synthesis added the prespecified glomerular GSE30528 compartment of source study GSE30122
14 and used restricted maximum likelihood with modified Hartung–Knapp inference, prediction
15 intervals, and Benjamini–Hochberg correction across all 145 genes. Complete three-study
16 estimates were available for 125/145 genes; none met meta-analysis $FDR < 0.05$ (0/145). In the
17 single external source study, 141/145 genes mapped, 94/141 followed the discovery direction,
18 and 39/145 met external Welch-test $FDR < 0.05$. Sample-label permutation, which preserved gene
19 correlation, supported 8/9 predefined pathways after pathway-level FDR correction; leukocyte
20 adhesion did not meet the threshold. These results support pathway-level transcriptional
21 concordance within one external glomerular study but do not establish independent multi-study
22 confirmation, causality, pathway activity, or cell-type localization. The analysis illustrates how

23 broader, selection-independent families and small-sample meta-analytic inference materially
24 alter conclusions drawn from a filtered gene subset.

25 **Keywords**

26 Diabetic kidney disease; complement; coagulation; extracellular matrix; transcriptomics; meta-
27 analysis; permutation testing

28 **Introduction**

29 Diabetic kidney disease (DKD) develops through interacting glomerular, tubular, vascular,
30 inflammatory, and fibrotic processes [1]. Human kidney transcriptomic studies have reported
31 complement-associated, immune, vascular, and extracellular-matrix expression changes across
32 DKD stages and tissue compartments [2–5]. Cross-study synthesis can help distinguish recurrent
33 signals from dataset-specific findings, but its inferential validity depends on keeping discovery,
34 filtering, and validation roles explicit.

35 A central difficulty arises when genes are selected using results from particular datasets and
36 those same datasets are then reused to calculate ordinary meta-analytic P values for the selected
37 subset. The resulting P values are conditional on an outcome-dependent filter and cannot be
38 interpreted as if the gene family had been fixed independently of the observed results. A second
39 difficulty occurs when two tissue compartments from the same parent study, including
40 measurements from shared donors, are treated as independent cohorts.

41 We addressed both issues by returning to the complete 145-gene union encoded by nine
42 biologically defined focused gene sets, before the outcome-dependent filter that retained 61
43 genes. Because the project lacks an external registration timestamp, this union is described as
44 analysis-defined rather than preregistered. We prespecified one GSE30122 compartment—

45 GSE30528 glomeruli—as the external estimand, used small-sample random-effects inference,
46 and evaluated pathway-level concordance with sample-label permutation. Single-nucleus
47 analyses were excluded from the primary claim because the available workflow lacked reference
48 mapping, doublet detection, expanded quality control, and a complete donor-level pseudobulk
49 analysis.

50 **Results**

51 **Analysis family and study roles**

52 The focused universe contained 145 unique genes distributed across nine overlapping biological
53 pathways. The earlier outcome-dependent workflow retained 61 genes using GSE142025
54 significance and GSE96804 directional consistency. In the present analysis, all 145 genes
55 remained in the inferential family. GSE142025 and GSE96804 defined discovery direction.
56 GSE30528, the glomerular compartment of source study GSE30122, was fixed as the external
57 compartment before the new result tables were inspected. GSE30529 was treated as a secondary
58 tubulointerstitial dataset from the same source study and was not entered as an additional
59 independent study (Fig. 1).

60 **Selection-independent three-study synthesis**

61 Complete Hedges' g estimates from GSE142025, GSE96804, and GSE30528 were available for
62 125/145 genes. Random-effects models used restricted maximum likelihood and modified
63 Hartung–Knapp t inference with two degrees of freedom. Non-estimable members were retained
64 as $P=1$ for correction over the complete 145-gene family. No gene met meta-analysis $FDR<0.05$
65 (0/145; Fig. 2 and Supplementary Table S1). The unmodified HKSJ intervals and both
66 unmodified and modified prediction intervals are supplied as sensitivities. Prediction intervals

were wide because only three source studies contributed, underscoring that a stable mean effect across future settings could not be established for individual genes.

Gene-level evaluation in one external source study

In GSE30528 glomeruli, 141/145 focused genes mapped to the platform and had estimable Welch tests. Of these, 94/141 followed the discovery direction, and 39/145 met $FDR < 0.05$ after correction over the complete focused family (Fig. 3 and Supplementary Table S2). These are external results relative to the GSE142025/GSE96804 discovery direction, but they derive from one source study and therefore do not establish replication across independent source studies.

Donor-disjoint tubulointerstitial sensitivity

Nine cross-accession donor pairs were inferred from sample titles and the shared GSE30122 parent study. Removing the corresponding GSE30529 samples left 5 DKD and 8 control tubulointerstitial samples. In this donor-disjoint sensitivity, 141/145 genes mapped, 86/141 followed the discovery direction, and 20/145 met within-compartment $FDR < 0.05$. Sample-label permutation supported 8/9 pathways after pathway-level FDR correction (Supplementary Tables S9–S11). These results demonstrate non-uniform gene-level behavior across compartments: 7 genes met $FDR < 0.05$ in both compartments, 32 only in GSE30528, and 13 only in donor-disjoint GSE30529. They do not convert the two compartments into independent studies.

Pathway-level sample-label permutation

To retain gene-gene correlation, disease labels were permuted among the 22 GSE30528 samples while preserving the 9-DKD versus 13-control allocation, and all genes were recomputed jointly in each of 10,000 permutations. Eight of nine pathways met pathway-level $FDR < 0.05$ (Fig. 4; Table 1). Complement activation, endothelial activation, macrophage recruitment, fibrosis,

89 extracellular-matrix remodeling, coagulation, hypoxia, and transforming-growth-factor-beta
90 signaling were supported. Leukocyte adhesion was directionally positive but did not meet the
91 adjusted threshold (FDR=0.0998).

92 **Unsupervised structure, outlier, and WGCNA audits**

93 PCA and sample-correlation diagnostics were run separately within each primary dataset. The
94 predefined median-correlation rule flagged 8 samples; because all flagged samples were DKD
95 samples and their biological versus technical origin could not be distinguished, they were
96 retained in the primary analysis. Excluding them in a sensitivity analysis did not change the
97 individual-gene conclusion (0/145 meta-analysis FDR<0.05; Supplementary Figs. S2–S3 and
98 Supplementary Tables S14–S16).

99 WGCNA was retained only as an internal GSE96804 description. Of the 145 focused genes, 58
100 mapped to the DKD-associated black module. Nineteen of these also met GSE30528 FDR<0.05,
101 whereas 39 black-module genes did not; conversely, 20 GSE30528 FDR-positive genes were
102 outside the black module or outside the WGCNA top-variance universe (Supplementary Table
103 S12). This partial overlap is compatible with compartment, cohort, variance-filter, and network-
104 construction differences and is not independent validation.

105 **Table 1. External pathway-level permutation results**

Pathway	Mapped genes	Aligned Hedges' g	Permutation P	BH FDR
COMPLEMENT_ACTIVATION	21	1.02	9.999e-05	0.00018
ENDOTHELIAL_ACTIVATION	16	0.83	9.999e-05	0.00018
MACROPHAGE_RECRUITMENT	16	0.8147	9.999e-05	0.00018

Pathway	Mapped genes	Aligned Hedges' g	Permutation P	BH FDR
FIBROSIS	12	1.073	9.999e-05	0.00018
ECM_REMODELING	15	0.6398	9.999e-05	0.00018
COAGULATION_CASCADE	21	0.4985	0.0003	0.00045
HYPOXIA	10	0.4689	0.0022	0.002828
TGF_BETA_SIGNALING	14	0.3593	0.0314	0.03532
LEUKOCYTE_ADHESION	11	0.2161	0.09979	0.09979

106 Discussion

107 The principal finding is narrower than the earlier filtered-subset interpretation. When the full
108 145-gene family was analyzed with REML and modified Hartung–Knapp inference, no
109 individual gene met meta-analysis FDR<0.05. The result does not imply that all underlying
110 expression differences are absent. It shows that the available three-study evidence is insufficient
111 for multiplicity-adjusted, small-sample random-effects claims at individual-gene resolution.
112 The independent evidence is stronger at pathway level within GSE30528. Eight predefined
113 pathways were concordant under a sample-label permutation that retained the observed cross-
114 gene structure. This supports an exploratory interpretation in which complement, coagulation,
115 vascular, inflammatory, hypoxia, transforming-growth-factor-beta, and extracellular-matrix
116 programs recur in the glomerular compartment of one external DKD study. Prior human kidney
117 studies provide biological context for these themes [2–5], but the present reanalysis does not
118 establish pathway activation, protein abundance, upstream regulation, or causality.

119 The contrast between 0/145 meta-analytic gene discoveries and 39/145 external Welch-test
120 discoveries is not contradictory. The former uses random-effects inference over only three

121 heterogeneous source studies with a t reference distribution and full-family multiplicity control.
122 The latter tests one external compartment and does not estimate between-study heterogeneity.
123 The pathway permutation then aggregates direction-aligned effects and gains stability while
124 preserving gene correlation. These estimands answer different questions and should not be
125 combined into a single “confirmed gene” count.

126 Model multiplicity was addressed by fixing one primary model—REML with modified Hartung–
127 Knapp inference—and one 145-gene correction family. Unmodified HKSJ intervals, prediction
128 intervals, sex adjustment, and correlation-outlier exclusion are labeled sensitivities and were not
129 searched to choose a favorable result. A Bonferroni division by the six historical meta-analysis
130 variants is therefore not applied to the new primary model; the historical variants no longer
131 constitute six co-primary opportunities for discovery.

132 GSE30528 and GSE30529 should not be presented as separate sources of validation. Both are
133 compartments of GSE30122, and nine cross-accession donor pairs can be inferred from sample
134 titles and the shared parent study. Restricting the primary external estimand to GSE30528 avoids
135 introducing an unestimated cross-compartment covariance. The earlier filtered 52-gene analysis
136 reported 39/52 direction concordance in donor-disjoint GSE30529 but did not meet its matched-
137 background criterion. That statistic is retained only as historical context because the candidate set
138 and null construction were outcome-dependent; it is not combined with the new 145-gene tests.

139 The new full-family sensitivity clarifies, but does not erase, the compartment difference. Donor
140 removal reduced GSE30529 from 10 DKD and 12 control samples to 5 and 8, respectively,
141 making single-gene estimates less precise and more sensitive to individual samples. The two
142 microdissected compartments also contain different cell mixtures and may carry genuinely
143 different DKD responses: glomerular endothelial, mesangial, and podocyte programs need not

144 track tubulointerstitial epithelial, stromal, vascular, and immune signals gene by gene. Manual
145 microdissection, residual vascular content, probe behavior, and incomplete clinical metadata
146 provide additional technical or compositional explanations. Consistent with this heterogeneity,
147 only 7 genes met full-family $FDR < 0.05$ in both compartments, whereas 32 were GSE30528-only
148 and 13 were donor-disjoint-GSE30529-only. At pathway level, however, 8/9 sets were supported
149 in each compartment, although the unsupported set differed. This pattern suggests that broad
150 biological programs may recur while their contributing genes and effect magnitudes vary by
151 compartment. It does not prove a compartment-specific mechanism, because donor pairing was
152 inferred from titles, covariates were unavailable, and the retained GSE30529 sample was small.
153 The correct conclusion is therefore same-study, compartment-sensitive concordance—not
154 universal DKD replication and not evidence from two independent validation cohorts.

155 Across the nine paired donors, the median gene-wise cross-compartment expression correlation
156 was only 0.219. This descriptive quantity is not the correlation between cohort-level disease-
157 effect estimates and therefore cannot identify the ρ parameter used in the retired correlated-
158 compartment meta-analysis. Rather than choosing ρ from these nine inferred pairs, the primary
159 analysis removed ρ dependence by representing GSE30122 with one prespecified compartment.
160 The former ρ -grid models are retained only as historical sensitivity analyses and are not used
161 to select genes or conclusions.

162 Single-nucleus results were removed from the title, abstract, primary results, and principal
163 conclusion. Existing marker-score labels may be shown separately as provisional methodological
164 context only. Cell-type localization should be reconsidered after external reference mapping,
165 doublet detection, expanded nucleus-level quality control, donor-aware integration, and donor-
166 level pseudobulk analysis [12,13].

167 **Methods**

168 **Focused universe and discovery boundary**

169 The analysis-defined universe was the union of nine pathway gene sets covering complement
170 activation, coagulation, endothelial activation, macrophage recruitment, leukocyte adhesion,
171 fibrosis, extracellular-matrix remodeling, hypoxia, and transforming-growth-factor-beta
172 signaling. The union contained 145 unique genes. Repository lineage placed this union before
173 the result-dependent 145-to-61 filter, but no external preregistration or trusted third-party
174 timestamp was available. All 145 genes were retained for multiplicity adjustment.
175 GSE142025 and GSE96804 were the discovery datasets. For each gene, discovery direction was
176 the sign of the inverse-variance fixed-effect mean of the two Hedges' g estimates. GSE30528
177 glomeruli were the prespecified external compartment from parent study GSE30122. GSE30529
178 tubulointerstitium was excluded from the primary estimator because it belongs to the same
179 source study and includes repeated donors.

180 **Expression contrasts and effect sizes**

181 Human kidney datasets were analyzed separately rather than merged across platforms.
182 GSE142025 compared 21 advanced-DKD samples with 9 controls. GSE96804 compared 41
183 DKD glomerular samples with 20 controls. GSE30528 compared 9 DKD with 13 control
184 glomerular samples. For GEO microarrays, platform gene symbols were uppercased and, when
185 multiple probes mapped to one gene, the probe with the highest mean expression across all
186 samples was selected without using disease labels. Each study-level effect was Hedges' g for the
187 case-minus-control standardized mean difference [8]. External gene-level P values used two-
188 sided Welch tests.

189 **Random-effects synthesis and multiplicity**

190 For each gene with all three effects estimable, between-study variance was estimated by
191 restricted maximum likelihood. The pooled effect used inverse total-variance weights. Because
192 only three studies contributed and their precisions differed, primary uncertainty used modified
193 Hartung–Knapp with scale $\max(1, q)$, a t reference distribution, and $k-1$ degrees of freedom;
194 unmodified HKSJ results were retained as a sensitivity [9,10]. Prediction intervals used $k-2$
195 degrees of freedom. Genes without complete three-study mapping were assigned $P=1$ for family-
196 wise bookkeeping. Benjamini–Hochberg correction was applied to exactly 145 P values [7].

197 **External gene and pathway evaluation**

198 External gene-level Welch P values in GSE30528 were corrected across all 145 genes, with
199 unmapped genes retained as $P=1$. Direction concordance compared the GSE30528 Hedges' g
200 sign with the discovery direction.
201 For pathway testing, the statistic was the mean discovery-direction-aligned Hedges' g among
202 mapped genes in each pathway. Disease labels were permuted 10,000 times among GSE30528
203 samples while retaining nine case labels. All genes were recomputed jointly for every
204 permutation, retaining gene correlation induced by the observed expression matrix. One-sided P
205 values used the plus-one correction, and the nine pathway P values were BH-adjusted.

206 **Donor-disjoint and diagnostic sensitivities**

207 Donor numbers were parsed from GSE30528 and GSE30529 sample titles. The nine numbers
208 appearing in both accessions were treated as inferred repeated donors. Those nine GSE30529
209 samples were removed before the donor-disjoint tubulointerstitial sensitivity, leaving 5 DKD and
210 8 control samples. Welch tests and BH correction were repeated across all 145 genes, and the

211 nine pathway statistics were recomputed in 10,000 joint sample-label permutations with a fixed
212 5-versus-8 allocation. Gene-wise Pearson correlations across the nine paired compartment
213 measurements were calculated descriptively; they were not treated as estimates of cross-study
214 effect covariance.

215 Within each primary dataset, PCA and Pearson sample-correlation matrices used the 1,000 genes
216 with highest within-dataset variance. Potential correlation outliers were defined before inspecting
217 downstream sensitivity results as samples with a robust z score below -3 for median correlation
218 to all other samples. Flagged samples were not automatically removed because disease biology
219 and technical artifacts could not be distinguished. The primary meta-analysis was repeated after
220 their exclusion as a sensitivity. GSE96804 additionally permitted an intercept-plus-disease-plus-
221 sex linear model; no common adjusted model was estimable across datasets.

222 WGCNA module membership was imported from the existing GSE96804 network analysis [11].
223 Overlap with the 145-gene universe and GSE30528 gene-level results was summarized
224 descriptively. Because the network and one discovery direction component both came from
225 GSE96804, no WGCNA P value was interpreted as external confirmation.

226 **Reproducibility**

227 The analysis archived the focused universe, per-study effects, discovery directions, complete
228 random-effects output, external gene tests, pathway permutation results, donor-disjoint
229 sensitivities, sample-level metadata, diagnostics, analysis settings, input hashes, and an output
230 manifest. The execution environment used Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, SciPy
231 1.16.3, statsmodels 0.14.5, and Matplotlib 3.10.6. M18 recomputed statistical results from frozen
232 processed matrices and cached normalized GEO SOFT values; it did not independently repeat

233 every upstream raw-read or CEL-file preprocessing step. A public immutable repository release
234 and archival DOI remain required before submission.

235 **Software and archival limitations**

236 Historical discovery-stage limma calls did not explicitly archive the trend and robust arguments
237 and therefore depended on the recorded package version's defaults. Some older Python stages
238 recorded major/minor but not patch versions. The prior single-nucleus phase did not archive a
239 final h5ad object and was reconstructed from intermediate files; those results are excluded from
240 the present claims. The present reanalysis records its full Python patch versions and does not use
241 the historical limma P values or reconstructed single-nucleus labels for primary inference. These
242 changes improve current reproducibility but do not retrospectively make every upstream
243 historical step raw-data reproducible.

244 **Limitations**

245 The 145-gene universe has internal pipeline lineage but no external registration timestamp; it
246 should not be described as preregistered. Only three independent source studies contributed to
247 the primary synthesis, making heterogeneity estimates and prediction intervals imprecise even
248 with small-sample methods. GSE30528 provides only one external source study, so its two renal
249 compartments cannot establish multiple-study replication. The paired-donor expression
250 correlations cannot identify correlation between compartment-level disease-effect estimates; the
251 primary model therefore avoids a rho parameter rather than claiming to estimate it.
252 Individual-level batch, age, treatment, renal-function, and other covariates were incomplete or
253 absent in the public annotations. GSE96804 recorded sex, but the other primary datasets did not
254 supply a common covariate set. Residual confounding and technical structure therefore cannot be

255 excluded. Correlation diagnostics flagged eight DKD samples, but these were not assumed to be
256 technical failures. Probe selection by highest all-sample mean was label-independent but may
257 favor highly expressed isoforms and does not resolve transcript-level effects. The primary
258 computations start from processed expression matrices or normalized GEO values, not a uniform
259 raw-data reprocessing pipeline.

260 The nine pathways overlap in gene membership. The sample-label permutation preserved within-
261 dataset gene correlation, but pathway-level BH correction did not provide a max-statistic family-
262 wise test across overlapping sets. The results are transcript-level associations and do not establish
263 protein abundance, pathway activity, molecular regulation, or causality.

264 The single-nucleus workflow available for this project lacked reference mapping, doublet
265 detection, expanded UMI and mitochondrial quality thresholds, donor-aware integration, and a
266 complete donor-level pseudobulk analysis. It therefore cannot support cell-type localization
267 claims in the present manuscript.

268 **Data availability**

269 All analyzed datasets are publicly available from the Gene Expression Omnibus:

- 270 • GSE142025: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE142025>
- 271 • GSE96804: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE96804>
- 272 • GSE30528: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30528>
- 273 • GSE30529: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30529>
- 274 • GSE166239: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE166239>
- 275 • GSE131882: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE131882>
- 276 • GSE111154: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE111154>

277 Processed effect-size tables, pathway-permutation results, and analysis manifests require a public
278 archival record with an immutable DOI before submission.

279 **Code availability**

280 Analysis code, environment specifications, source-data tables, and machine-readable manifests
281 require a public version-controlled release and a permanent archival DOI before submission.

282 **References**

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324 **Ethics statement**

325 This study used publicly available, de-identified human transcriptomic data. No new human or
326 animal samples were collected.

327 **Author contributions**

328 Lizhen Deng: Conceptualization, methodology, software, formal analysis, investigation, data
329 curation, visualization, writing—original draft, writing—review and editing, and project
330 administration.

331 **Competing interests**

332 The author declares no competing interests.

333 **AI disclosure**

334 OpenAI Codex was used under author supervision for code generation, comparison of statistical
335 options, statistical remediation, consistency checks, figure and table preparation, and language
336 drafting. Figures were generated by AI-assisted code from the archived numerical tables; no
337 generative image model was used. The author retains scholarly judgement and responsibility and
338 must independently verify the code, numerical outputs, citations, interpretation, and final
339 submitted text. AI was not used to generate or alter primary data.

340 **Figure legends**

341 **Figure 1. Analysis boundaries used for statistical remediation.**

342 The 145-gene analysis-defined universe was carried through discovery-direction estimation, the
343 three-study synthesis, external gene testing, and pathway permutation. The outcome-dependent
344 61-gene subset was not used as the primary inferential family. GSE30529 was retained only as a
345 secondary same-study compartment.

346 **Figure 2. Selection-independent three-study synthesis across the focused universe.**

347 Panel a shows pooled Hedges' g against the modified Hartung–Knapp P value for genes with
348 complete estimates in GSE142025, GSE96804, and GSE30528. Labels identify the ten smallest
349 unadjusted P values for display only. Panel b shows every study effect, normal 95% confidence
350 interval, random-effects weight, and the modified Hartung–Knapp pooled interval for eight fixed
351 legacy display genes. These genes are not a separate inferential family. No gene met BH
352 FDR<0.05 across the 145-gene family.

353 **Figure 3. Gene-level evaluation in GSE30528 glomeruli.**

354 Discovery effects combine GSE142025 and GSE96804 without GSE30528. External effects are
355 Hedges' g values from GSE30528. Blue points are direction-concordant genes; dark blue
356 additionally meet external FDR<0.05; red points are direction-discordant. GSE30528 is one
357 external source study.

358 **Figure 4. Pathway-level sample-label permutation in GSE30528.**

359 Bars show the mean discovery-direction-aligned Hedges' g for each pathway. Labels report BH
360 FDR over nine pathways. Disease labels were permuted 10,000 times with the 9-versus-13
361 allocation preserved and all genes recomputed jointly.

362 **Supplementary legend**

363 **Supplementary Figure S1. Complete three-study estimates for 125 mapped genes.**

364 Points show REML pooled Hedges' g and modified Hartung–Knapp 95% confidence intervals.
365 The figure is descriptive; multiplicity-adjusted results are supplied in Supplementary Table S1.

366 **Supplementary Figure S2. Unsupervised PCA diagnostics.**

367 PCA used the 1,000 most variable genes separately within each primary dataset. Group
368 separation can reflect disease biology, composition, confounding, or unrecorded technical
369 structure.

370 **Supplementary Figure S3. Within-dataset sample-correlation diagnostics.**

371 Heatmaps show Pearson correlations computed from the 1,000 most variable genes within each
372 primary dataset. The predefined robust-z outlier audit is reported in Supplementary Table S15.